

BHARGAVI ARYA SIVA

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Data and AI Engineer with almost 3.9 Years experience in building analytics and artificial intelligence solutions across healthcare, banking, insurance, and airline domains. Experienced in fraud detection, claims analysis, and risk analytics, as well as developing LLM-based chatbots and machine learning solutions for real-world business problems. Strong in Python, SQL, and data platforms, with a consistent focus on data quality, compliance, and production-ready AI systems.

SKILLS & ABILITIES

- **Data Analysis & Visualization:** SQL, Power BI, Tableau, Excel, KPI Dashboards, Data Storytelling
- **Big Data & Processing:** Hadoop, Apache Spark, PySpark
- **Programming:** Python, SAS, MATLAB, Flask,
- **Statistics & Analytics:** Regression, Forecasting, Hypothesis Testing, A/B Testing, Clustering
- **Data Engineering:** ETL, Data Cleaning, Automation (Python), AWS, Airflow, Azure, FX Data Warehouse, API Integration
- **Machine Learning:** Scikit-learn, TensorFlow, PyTorch, LangChain , Random Forest, GBM
- **Domain Knowledge:** Fraud Detection, Risk Analytics, Healthcare & Financial Data

WORK EXPERIENCE

Agency for Health care Administration(AHCA)

System Project Consultant(Medicaid Program Integrity – Data Detection)

Jan 2026 - present
Florida, USA

- Perform statewide Medicaid fraud detection analysis using DBVisualizer and advanced SQL, querying large-scale claims datasets stored in Amazon Redshift to identify abuse, waste, and non-compliant billing patterns.
- Extract, integrate, and validate provider and claims data from FX Enterprise EDW (Enterprise Data Warehouse), Amazon Redshift, and PostgreSQL to support enterprise-level Program Integrity investigations.
- Design and execute complex SQL workflows to detect anomalies such as upcoding, duplicate billing, excessive procedures, and abnormal healthcare utilization trends across Florida Medicaid programs.
- Develop Python-based automation scripts and analytical pipelines in Visual Studio, enabling repeatable ETL-style data processing, validation, and fraud detection workflows.
- Developed an internal AI-powered assistant using OLAMA to support analysts with SQL query generation, data exploration, and investigative insights, improving productivity and accessibility to Medicaid data.

M Result Services

Software Development Engineer

Jun 2023 - Dec 2023
Hyderabad, India

- Prepared business requirements documentation and collaborated with cross-functional stakeholders to define data needs and reporting specifications.
- Developed and optimized LLM-based chatbots for airline, banking, and insurance domains using Kore.AI, improving query accuracy and user engagement.
- Prepared internal documentation and training materials for chatbot deployment teams and QA testers.
- Collaborated with the testing team to identify and resolve bot issues through log analysis, performance evaluation, and real-time debugging.
- Partnered with clients and business teams to define chatbot requirements, review analytical reports, and implement enhancements based on feedback.
- Created KPI dashboards to monitor bot accuracy, escalation rate, and operational performance, ensuring continuous improvement across multiple deployments.

Quadratic Insights Pvt. Ltd

Machine Learning Engineer

Credit & Debit Card Fraud Detection

Jan 2022 - Dec 2022
Hyderabad, India

- Designed and implemented large-scale fraud detection systems using Hadoop, PySpark, SQL, and Python, processing 8+ years of historical transaction data to predict in under 30 seconds whether a POS, credit, or debit card transaction was fraudulent, significantly reducing false alerts and financial loss.
- Developed a biweekly model retraining pipeline leveraging Python machine learning models such as Random Forest and Gradient Boosting (GDM), utilizing distributed processing on Hadoop for continuous model improvement.
- Built end-to-end ETL and data processing workflows using PySpark, SQL, AWS Glue, and Apache Airflow, automating data extraction, transformation, validation, and feature engineering across high-volume financial datasets.
- Conducted advanced EDA and analytics on domestic and international transaction data to uncover risk indicators and abnormal spending patterns, and created Power BI dashboards to monitor fraud detection rates and exposure metrics.
- Provided production support for enterprise data pipelines, including monitoring Airflow jobs, troubleshooting failures, optimizing SQL queries, and ensuring operational stability and real-time data availability.

Dabur India - Easypolicy & Capitalgram Insurance Brokers
Machine Learning Engineer

Mar 2020 - Dec 2021
Noida, India

- Built a multi-PDF OCR + NLP engine to extract customer, policy, and financial details from thousands of insurance documents across multiple carriers, consolidating them into structured Excel models for analysis.
- Automated data extraction, transformation, and aggregation workflows using Python, reducing manual processing time and improving data integrity in high-volume financial reporting.
- Created automated reporting and validation tools to support underwriting and due diligence efforts, providing accurate inputs for credit assessments and cash flow projections.
- Developed Excel-based portfolio models to analyze policy risk exposure, premium revenue forecasts, and operational cost savings from automation.

Spyry Techonologies
Cyber Security Intern

June 2018 - Jul 2018
Bangalore, India

- Conducted Vulnerability Assessment & Penetration Testing (VAPT) of web applications, analyzing security risks, OWASP Top 10 vulnerabilities, and practicing common attacks using Kali Linux.

ACADEMIC PROJECTS

Predicting Monthly Revenue from Banking Customers - Regression Modeling

Mar2025-Apr 2025

- Built regression models to forecast revenue based on user behavior, product usage patterns, and demographic features, enabling more targeted cross-sell strategies.
- Applied feature engineering, VIF analysis, and model selection using AIC/BIC.
- Proposed future improvements for fraud detection and BI integration.
- Compared model performance using Linear Regression, Random Forest, and LightGBM, achieved best RMSE with LightGBM and advanced feature selection.
- Documented analytical process flow, validation steps, and visual summaries to communicate results to non-technical stakeholders.

Credit Risk Modeling & Monitoring (May 2025 – Present)

- Built and compared Logistic Regression, Random Forest, Naïve Bayes, and Deep Neural Network (DNN) models to predict loan default probability using the Home Credit Default Risk dataset (300K + records).
- Conducted extensive data cleaning, preprocessing, and EDA handled missing values, detected outliers, and visualized feature distributions with Seaborn and Matplotlib.
- Merged multiple related datasets (bureau, credit card, installments, previous applications) using Python to create a unified analytical view.
- Applied feature engineering and PCA to reduce dimensionality while retaining 99 % of variance.
- Evaluated models using AUC-ROC, F1-Score, Precision, and Recall, identifying key drivers of default risk.
- Presented findings through data visualizations and a Power BI-style summary to support risk-aware lending decisions.

CERTIFICATIONS

- Tableau Desktop Specialist (TTD101) – Salesforce, 2026
- Microsoft AI-900: Azure AI and Cognitive Services
- Power BI, ML Ops, Pursuing certifications in LLMs, LangChain, Generative AI
- Pursuing certification in Tableau TTD201 Intermediate Certification in Salesforce

EDUCATION

Florida State University

Jan 2024 - Dec 2025

Master of Science, Statistical Data Science

GPA: 3.7/4.0

Coursework: Applied Regression, Time Series Analysis, Computational Methods, Advanced SAS, Machine Learning, A/B Testing, Experimental Design Advanced Machine Learning, Advanced Data Science, Hypothesis Testing.

CDAC - Center for Development of Advanced Computing, India

Aug 2019 - Feb 2020

Post-Graduate Diploma, Artificial Intelligence

Osmania University, India

Aug 2015 - Jun 2019

Bachelor of Engineering, Electronics & Communication Engineering